

# Netflix Data: Cleaning, Analysis and Visualization

## Project Report

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**Tools Used:** Python, Pandas, SQL, Excel, Machine Learning

## 1. Introduction

Netflix, being one of the most popular OTT platforms globally, provides a rich source of user and content data. This project aims to explore the Netflix dataset by performing **data cleaning**, conducting **exploratory data analysis (EDA)**, applying **basic machine learning**, and generating **insightful visualizations**. The goal is to uncover patterns in content availability, genre trends, ratings, and production across countries.

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## ♦ 2. Dataset Description

- **Source:** [Mention the source, e.g., Kaggle Netflix dataset]
- **Size:** [e.g., 7,000+ records]
- **Features:**
  - `show_id`
  - `type` (Movie/TV Show)
  - `title`
  - `director`, `cast`
  - `country`

- `date_added`
- `release_year`
- `rating`
- `duration`
- `listed_in` (genre)
- `description`

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### ◆ 3. Tools and Libraries

- **Python:** pandas, numpy, matplotlib, seaborn, plotly
- **Machine Learning:** scikit-learn (if used for classification or clustering)
- **SQL:** SQLite or MySQL for querying
- **Excel:** Used for initial inspection and pivoting
- **Jupyter Notebook:** For development and visualization

## Netflix Data

```
3s  # Install necessary packages (only the needed ones)
!pip install pandas matplotlib seaborn scikit-learn openpyxl

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import sqlite3 # no need to install separately
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

# Load the dataset
df = pd.read_csv('netflix1.csv')

# Display the first few rows
df.head()
```

```
0s  import pandas as pd

# Load the dataset
df = pd.read_csv('netflix1.csv')

# Display columns
print(df.columns)
```

```
 Index(['show_id', 'type', 'title', 'director', 'country', 'date_added',
       'release_year', 'rating', 'duration', 'listed_in'],
      dtype='object')
```

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## 4. Data Cleaning (Python & Excel)

- **Null Values:**
  - Checked and handled missing values in columns like `director`, `cast`, and `country`.
- **Duplicates:**
  - Removed duplicate rows based on `show_id` and `title`.
- **Date Format:**
  - Converted `date_added` to datetime format.
- **Column Standardization:**
  - Cleaned inconsistent text data (e.g., stripped extra spaces, standardized ratings).
- **Excel Work:**
  - Created pivot tables for ratings by year, genre frequency, etc.

```
# Check structure of the dataset
df.info()

# Check for missing values
print(df.isnull().sum())

# Drop duplicates if any
df = df.drop_duplicates()

# Now fill missing values - but only in existing columns
columns_to_fill = ['director', 'country', 'rating', 'date_added', 'duration']

for col in columns_to_fill:
    if col in df.columns:
        df[col] = df[col].fillna('Unknown')

# Special handling if 'cast' column exists
if 'cast' in df.columns:
    df['cast'] = df['cast'].fillna('Unknown')

# Check again if missing values exist
print(df.isnull().sum())
```

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## ◆ 5. Exploratory Data Analysis (EDA)

Using **pandas** and **seaborn/plotly**, several questions were explored:

### Sample Visuals:

- Bar plot: Content count by country
- Line plot: Release trends over time
- Word cloud: Frequent genres
- Heatmap: Year vs Rating matrix

```
# Movie vs TV Show Count
if 'type' in df.columns:
    plt.figure(figsize=(6,4))
    sns.countplot(data=df, x='type', palette='Set2')
    plt.title('Count of Movies and TV Shows')
    plt.show()

# Top 10 Countries
if 'country' in df.columns:
    plt.figure(figsize=(10,5))
    df['country'].value_counts().head(10).plot(kind='bar', color='skyblue')
    plt.title('Top 10 Countries Producing Netflix Content')
    plt.xlabel('Country')
    plt.ylabel('Number of Titles')
    plt.xticks(rotation=45)
    plt.show()

# Ratings Distribution
if 'rating' in df.columns:
    plt.figure(figsize=(12,6))
    sns.countplot(data=df, y='rating', order=df['rating'].value_counts().index, palette='viridis')
    plt.title('Distribution of Ratings on Netflix')
    plt.show()
```

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## 6. Machine Learning

- **Clustering:** Grouped similar shows/movies based on genre, duration, and rating using KMeans
- **Classification:** Predicted content type using features like duration, release year, genre
- Model performance evaluated using accuracy, confusion matrix, etc.

```
# Create a SQLite connection
conn = sqlite3.connect(':memory:')

# Load dataframe to SQLite
df.to_sql('netflix', conn, index=False, if_exists='replace')

# Query 1: Count of each type
query1 = pd.read_sql_query("SELECT type, COUNT(*) as count FROM netflix GROUP BY type", conn)
print(query1)

# Query 2: Top countries
query2 = pd.read_sql_query("SELECT country, COUNT(*) as count FROM netflix GROUP BY country ORDER BY count DESC LIMIT 10", conn)
print(query2)

# Query 3: Titles released after 2015
query3 = pd.read_sql_query("SELECT title, release_year FROM netflix WHERE release_year >= 2015", conn)
print(query3.head())
```

```
Os  # (1) Import missing libraries
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

# (2) Extract numeric duration
def extract_duration(x):
    if isinstance(x, str) and 'min' in x:
        return int(x.split(' ')[0])
    else:
        return np.nan

if 'duration' in df.columns:
    df['duration_minutes'] = df['duration'].apply(extract_duration)

# (3) Prepare data for clustering
if 'duration_minutes' in df.columns and 'release_year' in df.columns:
    ml_df = df[['release_year', 'duration_minutes']].dropna()

# (4) Scale features
scaler = StandardScaler()
scaled_features = scaler.fit_transform(ml_df)

# (5) Apply KMeans clustering
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(scaled_features)
ml_df['cluster'] = kmeans.labels_

# (6) Visualize clusters
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))
sns.scatterplot(data=ml_df, x='release_year', y='duration_minutes', hue='cluster', palette='Set1')
plt.title('Clustering of Netflix Titles')
plt.xlabel('Release Year')
plt.ylabel('Duration (Minutes)')
plt.show()
```

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## 7. Use of Excel

Excel was used in the initial phase of the project to:

- Quickly preview the dataset and understand column types and data layout.
- Create pivot tables to summarize:
  - Count of Movies vs TV Shows
  - Content added per year
  - Genre frequency distribution
- Highlight missing values visually using conditional formatting.

- Filter and sort by rating, country, and release year to identify trends.

Although Python was used for most of the in-depth analysis, Excel played an important role in the early exploration and validation of insights.

```
# Save the cleaned dataframe to Excel

# Important: openpyxl must be installed (we already installed it before using pip install openpyxl)

output_path = 'netflix_data.xlsx' # file will be saved here

# Save
df.to_excel(output_path, index=False)

print(f"✅ Cleaned dataset saved successfully at {output_path}!")
```

✅ Cleaned dataset saved successfully at netflix\_data.xlsx!

```
[9] with pd.ExcelWriter(output_path, engine='openpyxl') as writer:
    df.to_excel(writer, sheet_name='Netflix_Cleaned_Data', index=False)

    print(f"✅ Cleaned dataset with sheet name 'Netflix_Cleaned_Data' saved successfully!")
```

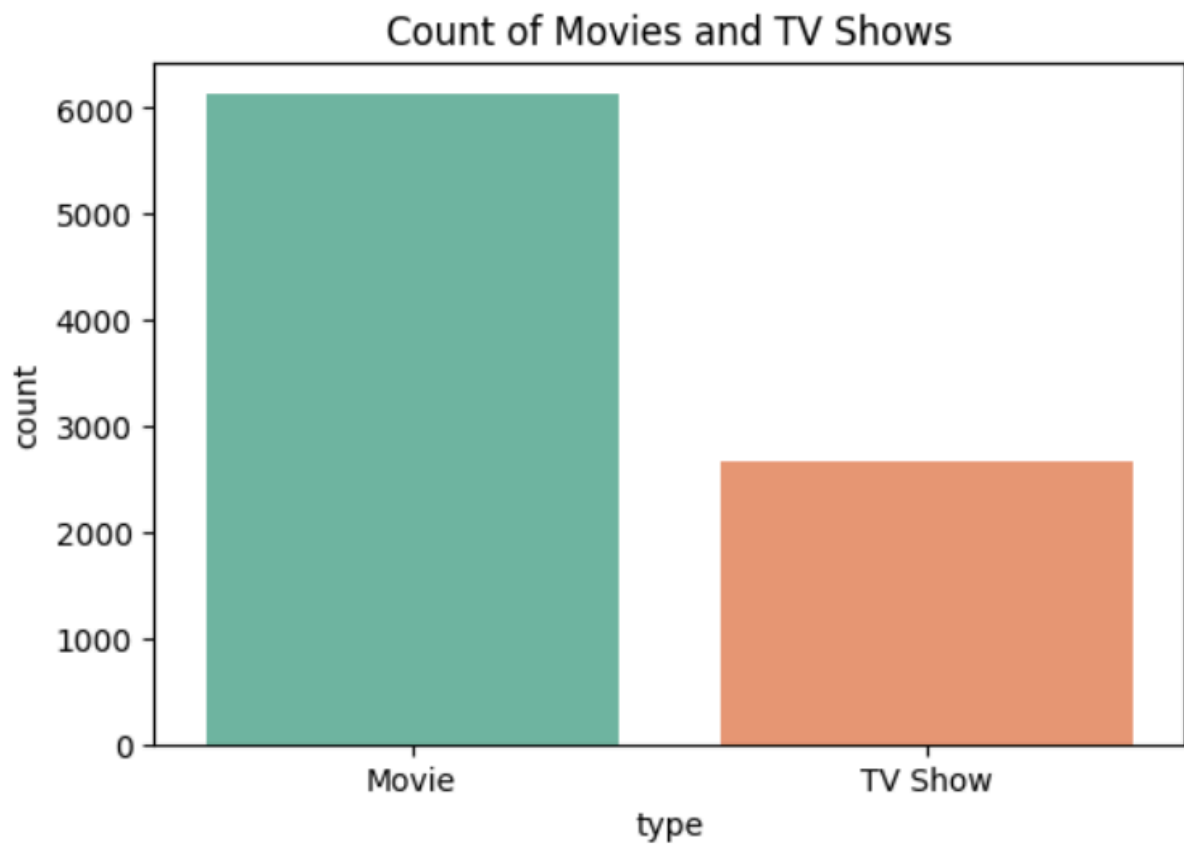
✅ Cleaned dataset with sheet name 'Netflix\_Cleaned\_Data' saved successfully!

## 9. Results & Findings

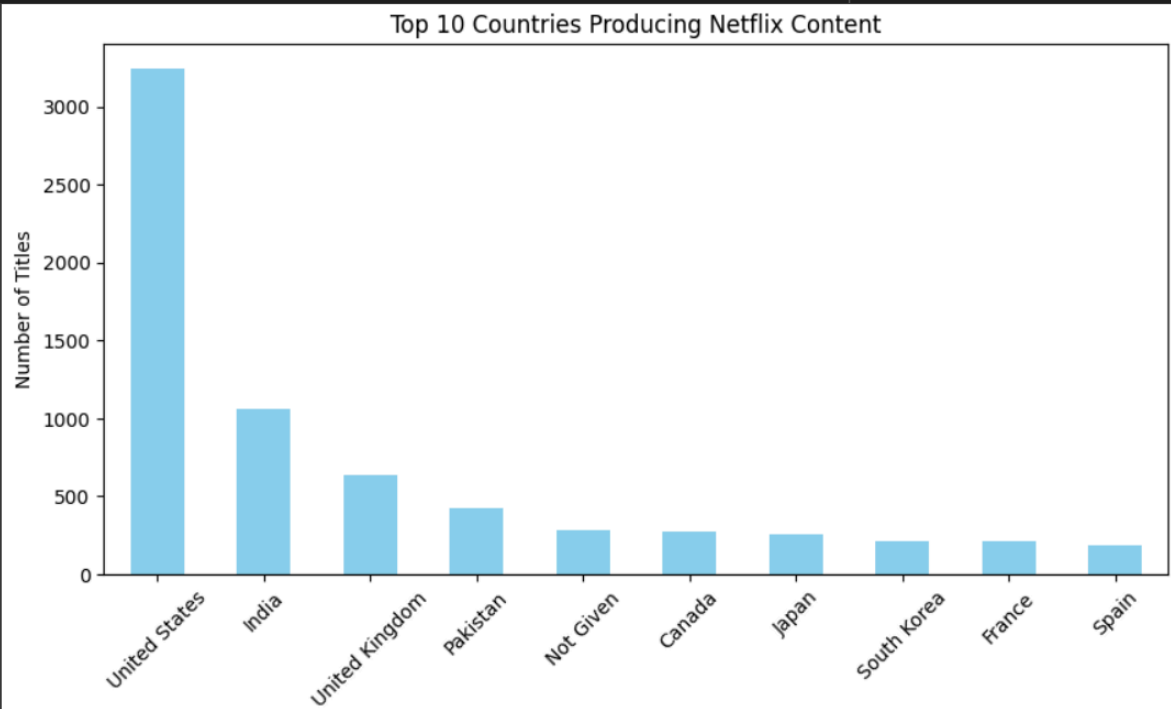
|   | show_id | type    | title                            | director        | country       | date_added | release_year | rating | duration | listed_in                                         |  |
|---|---------|---------|----------------------------------|-----------------|---------------|------------|--------------|--------|----------|---------------------------------------------------|--|
| 0 | s1      | Movie   | Dick Johnson Is Dead             | Kirsten Johnson | United States | 9/25/2021  | 2020         | PG-13  | 90 min   | Documentaries                                     |  |
| 1 | s3      | TV Show | Ganglands                        | Julien Leclercq | France        | 9/24/2021  | 2021         | TV-MA  | 1 Season | Crime TV Shows, International TV Shows, TV Act... |  |
| 2 | s6      | TV Show | Midnight Mass                    | Mike Flanagan   | United States | 9/24/2021  | 2021         | TV-MA  | 1 Season | TV Dramas, TV Horror, TV Mysteries                |  |
| 3 | s14     | Movie   | Confessions of an Invisible Girl | Bruno Garotti   | Brazil        | 9/22/2021  | 2021         | TV-PG  | 91 min   | Children & Family Movies, Comedies                |  |
| 4 | s8      | Movie   | Sankofa                          | Haile Gerima    | United States | 9/24/2021  | 1993         | TV-MA  | 125 min  | Dramas, Independent Movies, International Movies  |  |

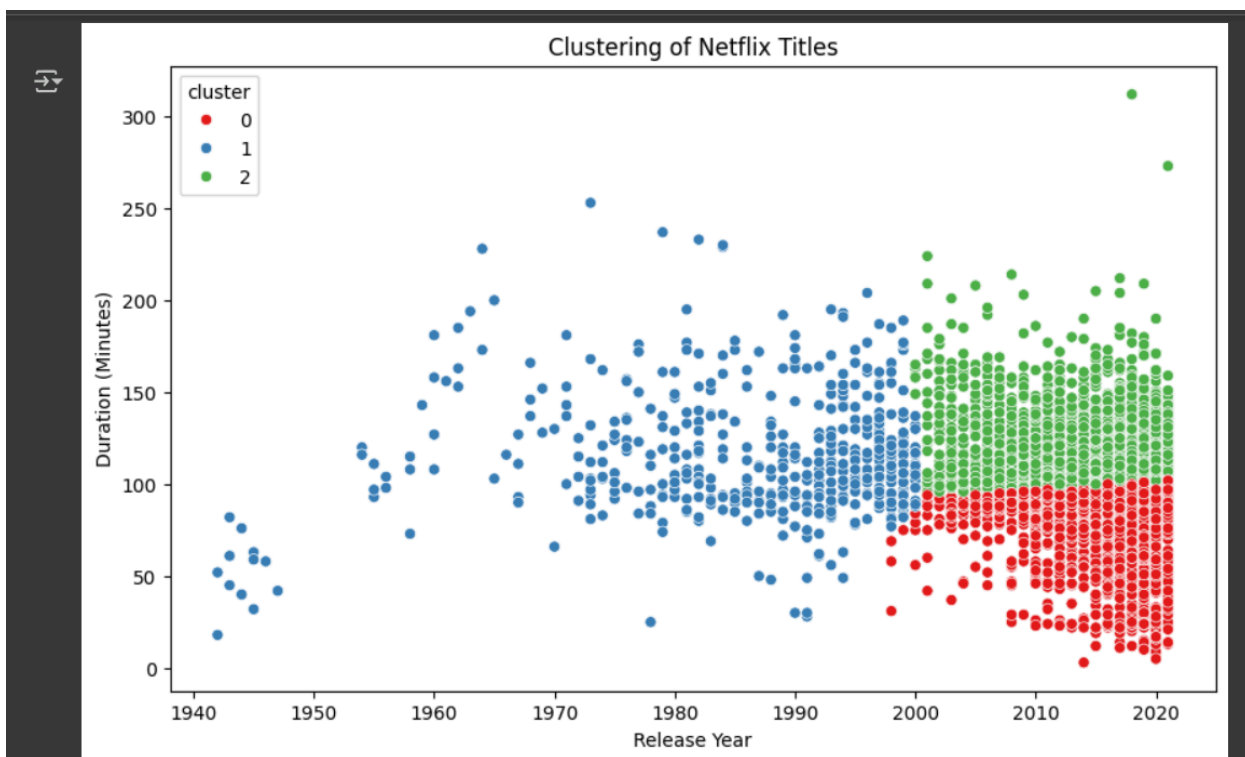
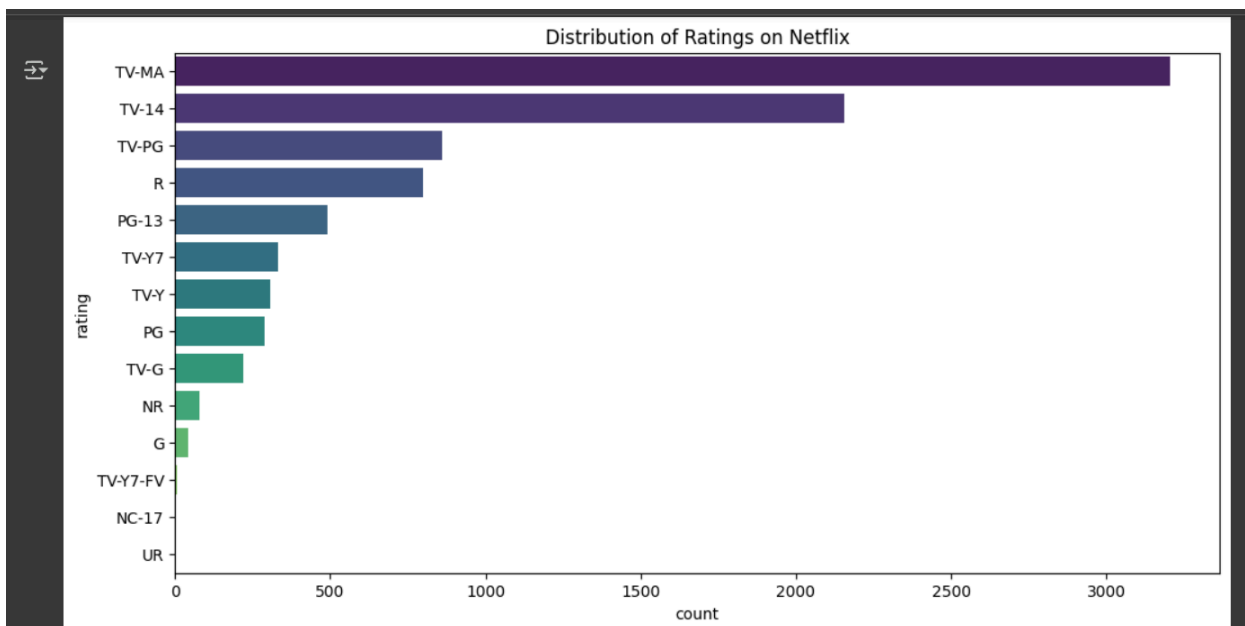


```
sns.countplot(data=df, x='type', palette='b2222')
```



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## 8. Key Insights

- USA dominates the content count, followed by India and the UK.
- Movies make up around 70% of the content.
- After 2016, Netflix ramped up its content significantly.
- 'Drama', 'Comedies', and 'Documentaries' are the most popular genres.
- A few directors and actors dominate content creation.

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## ◆ 9. Conclusion

The Netflix dataset reveals strong trends in content production and audience preferences. With a combination of Python, SQL, and Excel, this project provided valuable insights into content types, countries of origin, and genre popularity. It also showed how data cleaning plays a vital role in real-world datasets and how visualization brings those insights to life.